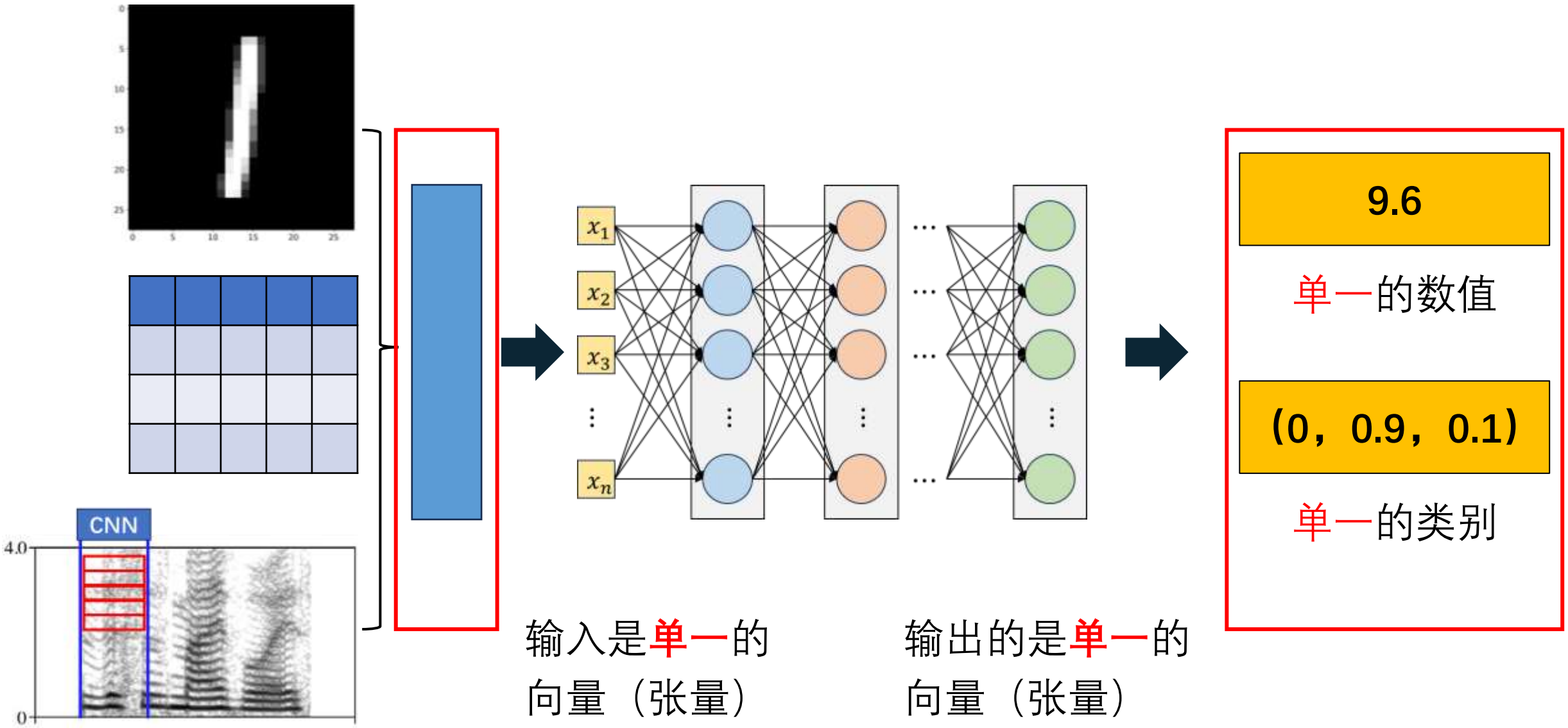
第7.2节 序列到序列问题

Sequence-to-sequence & RNN & Attention

Class-to-Class



输入是单一的向量（张量）

输出的是单一的向量（张量）

除了Class-to-Class，还有这么些实际问题……

“There is only one truth”

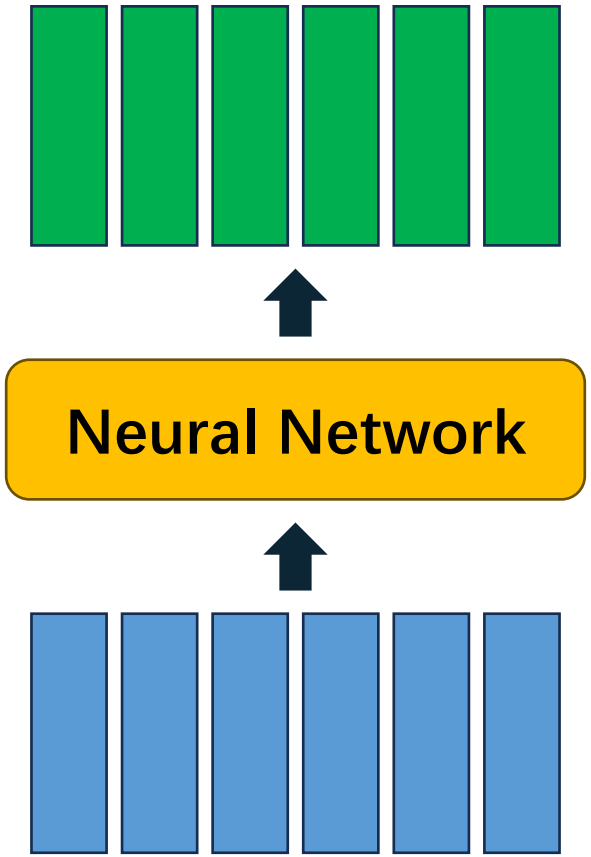
“真相只有一个”

“To go to the moon, and stay there, and do the other things we've talked about, as one of the great achievements of the 20th century.”

Neural Network

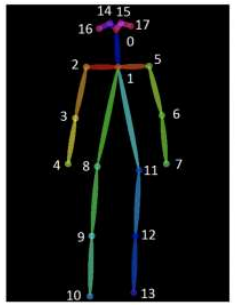
“真実はいつもひとつ”

序列到序列



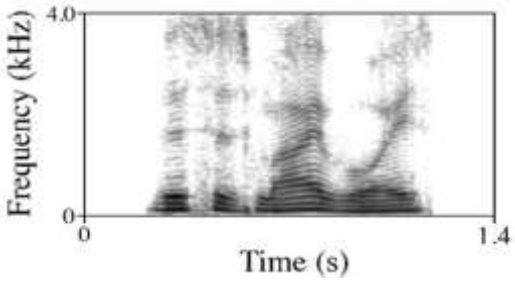
“There is only one truth”

“真相只有一个”

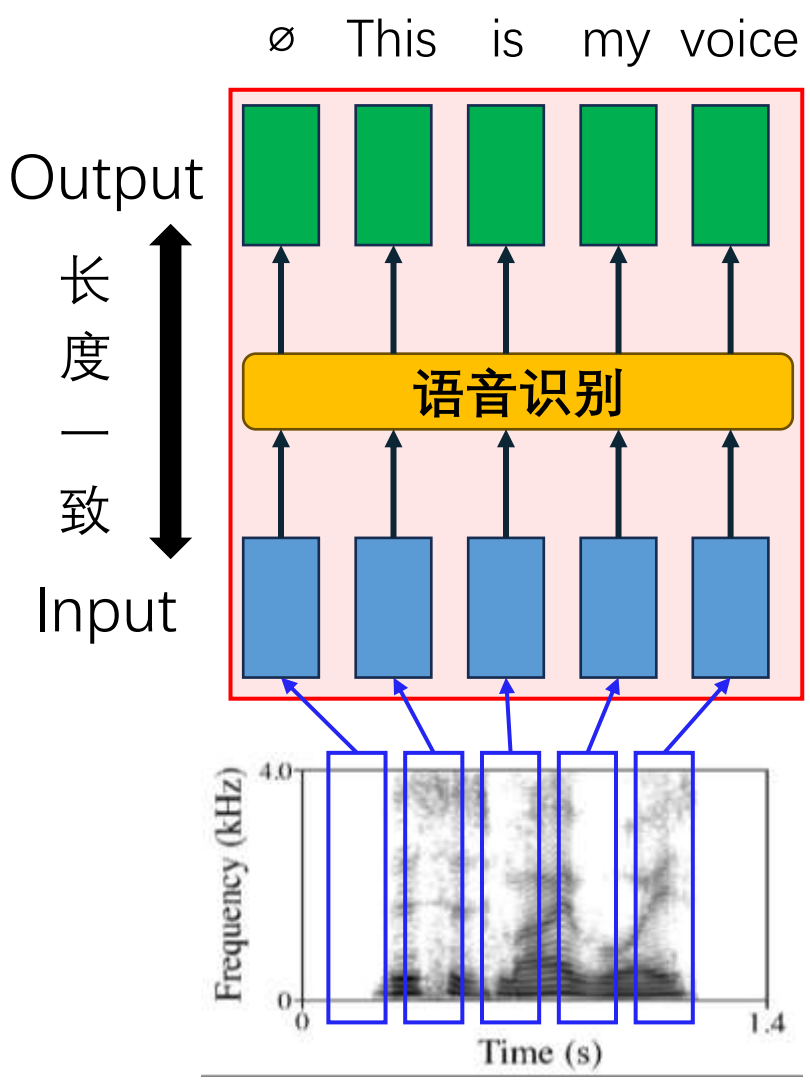


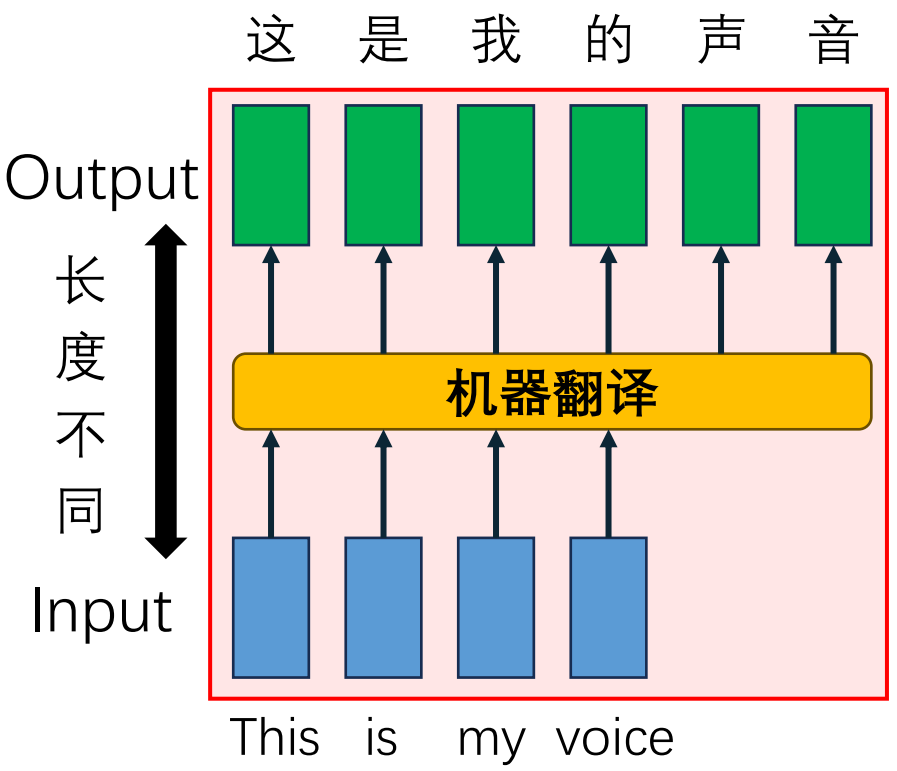
输入和输出皆是序列

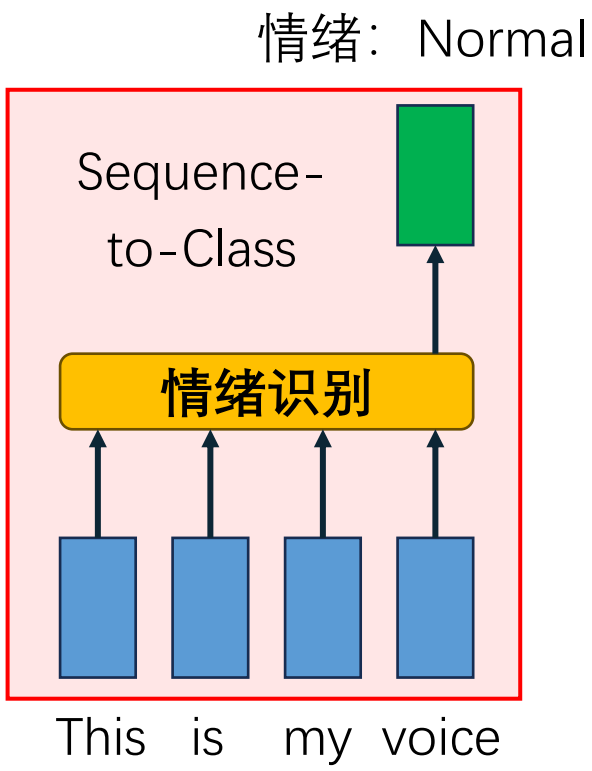
“真実はいつ
もひとつ”



序列到序列：问题分类



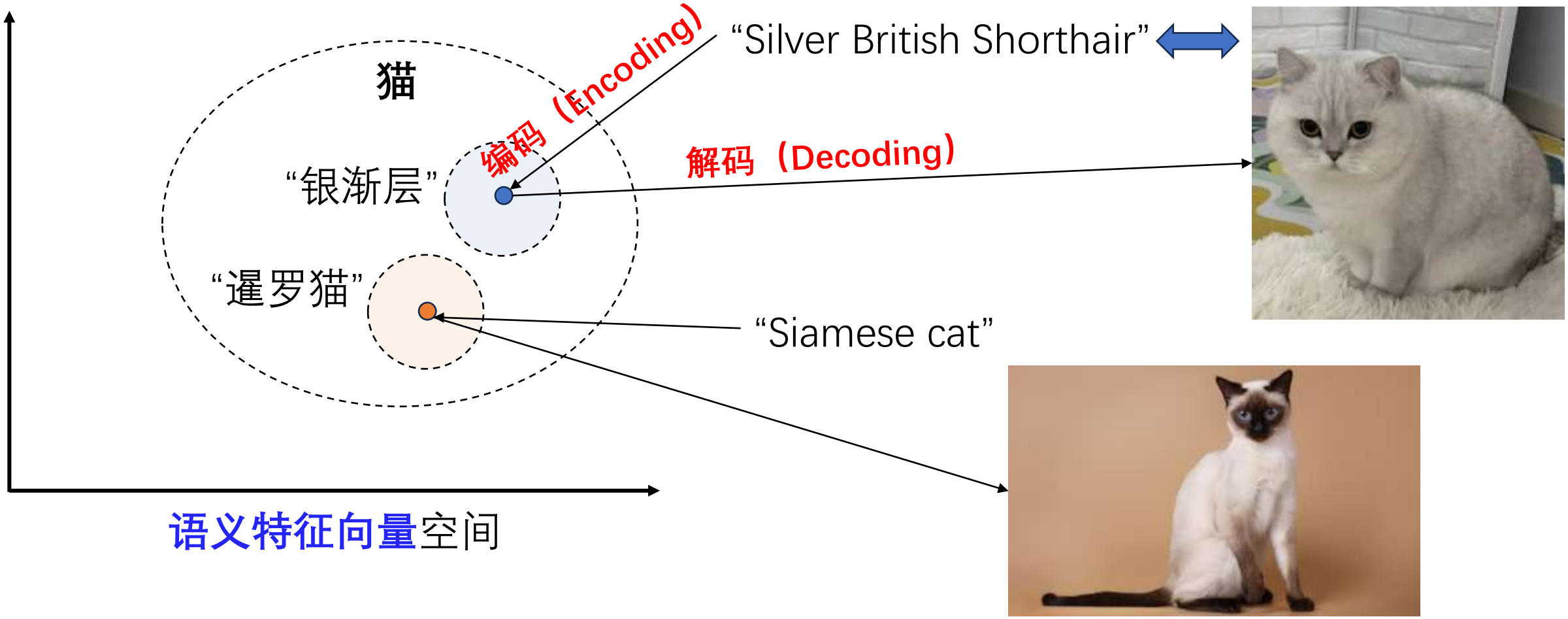




核心问题依然是“**编码-解码**”

序列到序列：实现过程

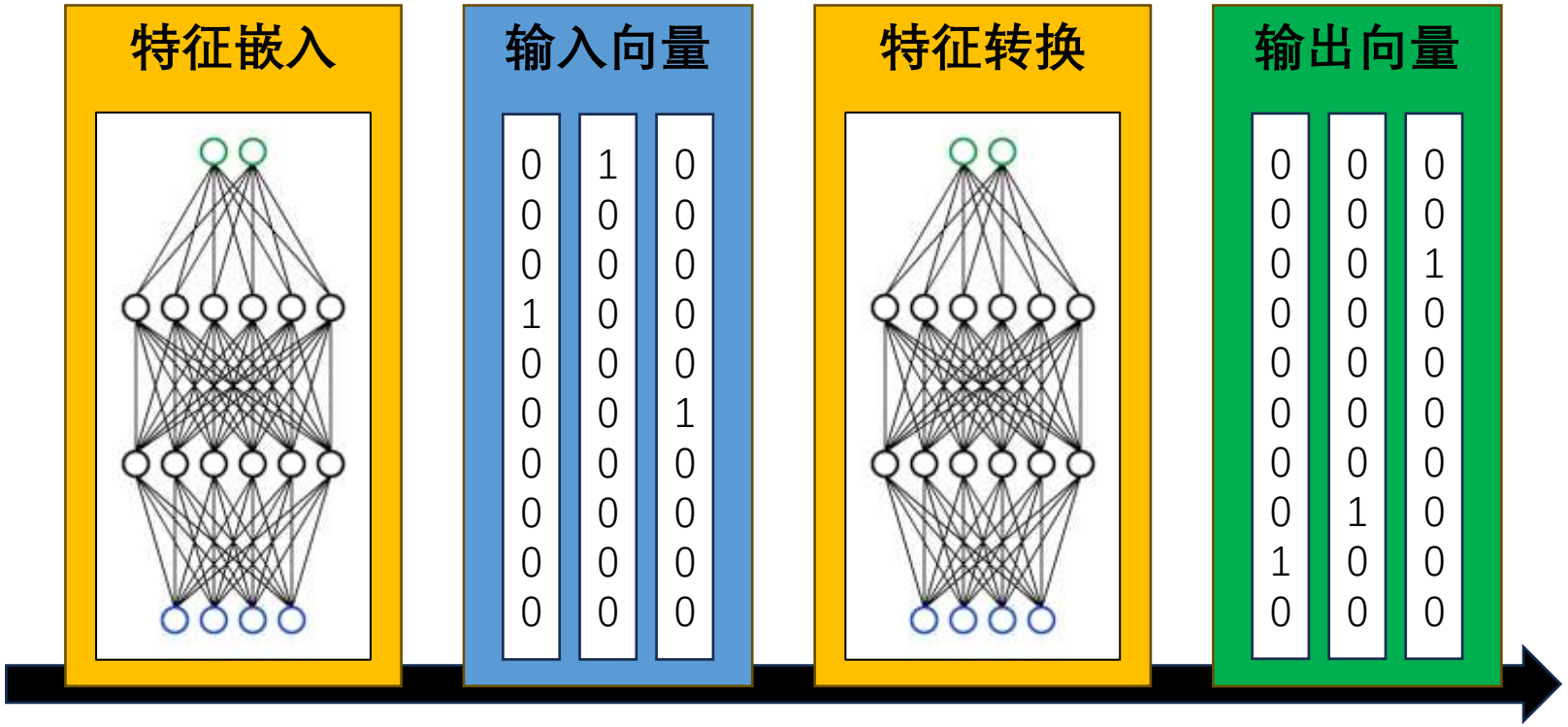
基于语义空间的编码-解码假设



序列到序列：实现过程

结合“语义空间的编码-解码假设”，思考如何实现**序列到序列**的问题

“真実はいつもひとつ”



“真相只有一个”

“There is only one truth”

投影至统一特征空间　　构建输入到输出的映射

编码（Encoding）　　**解码（Decoding）**

问题1：如何进行编码（Encoding）？以机器问答为例……

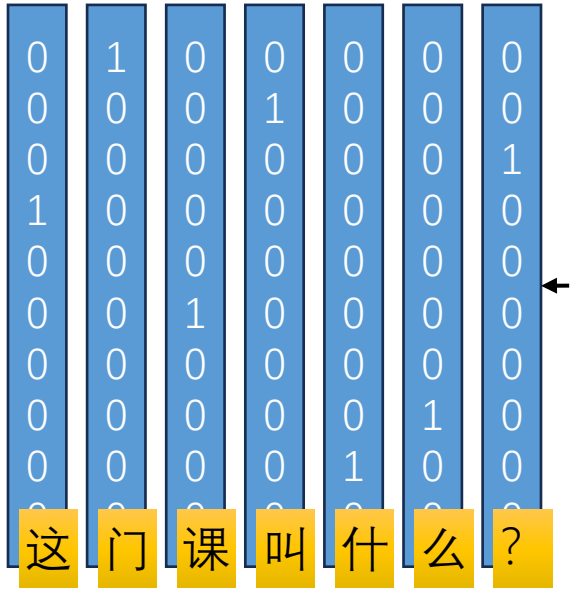
这门课叫什么？

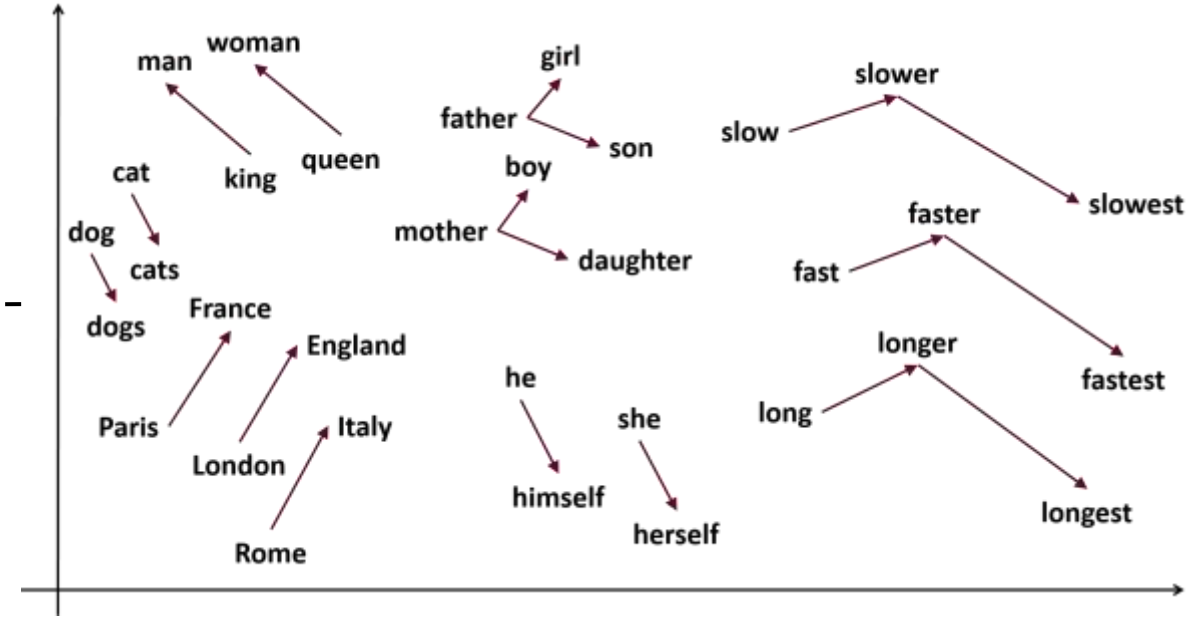
机器学习

在Embedding的空间中，词与词之之间的距离在一定程度上反映出了其本质的意义与关系

“这门课叫什么？” ⇨ {这, 门, 课, 叫, 什, 么, ？} ⇨ (0,0,0,1,0,0,0,0,0,0)……

⬇向量表示（词向量） 词分器 Tokenizer　　词嵌入 Embedding





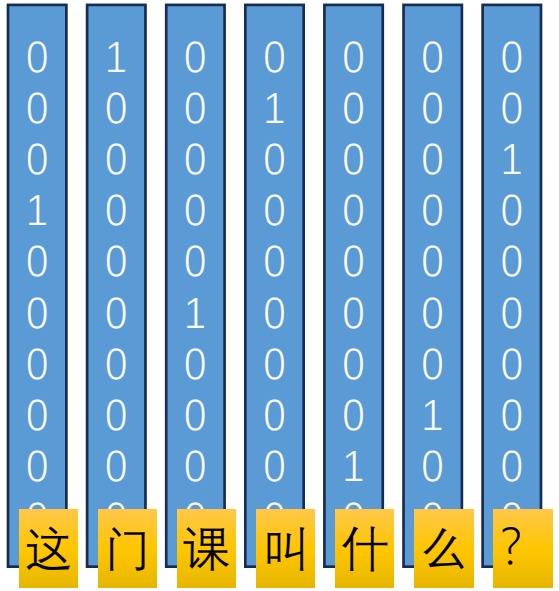
问题2：如何进行解码（Decoding）？

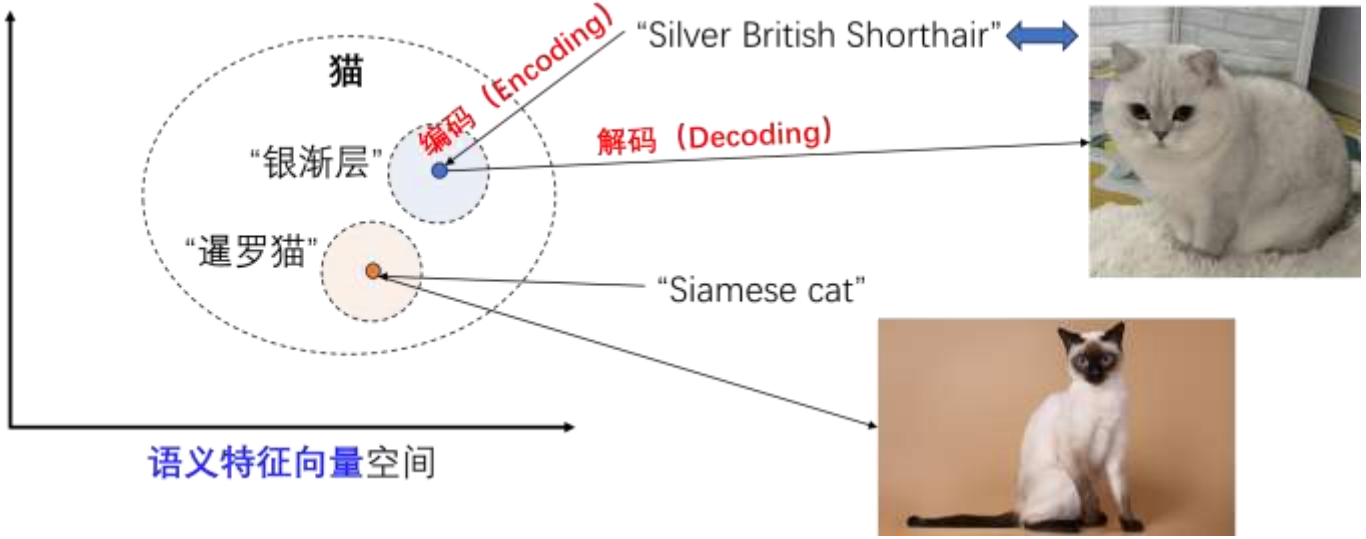
这门课叫什么？

机器学习

“这门课叫什么？” ⇨ “机器学习”

输入与输出语义不同





问题2：如何进行解码（Decoding）？

这门课叫什么？

机器学习

“这门课叫什么？”

用前文的语义编码解码（生成）后文的内容

几乎所有涉及解码任务都可以转化为生成问题

问题2：如何进行解码（Decoding）？

机器翻译:

“问：真実はいつもひとつ” “答：真”

“问：真実はいつもひとつ真” “答：相”

“问：真実はいつもひとつ真相” “答：只”

“问：真実はいつもひとつ真相只” “答：有”

“……”

机器问答:

“问：这门课叫什么？” “答：机”

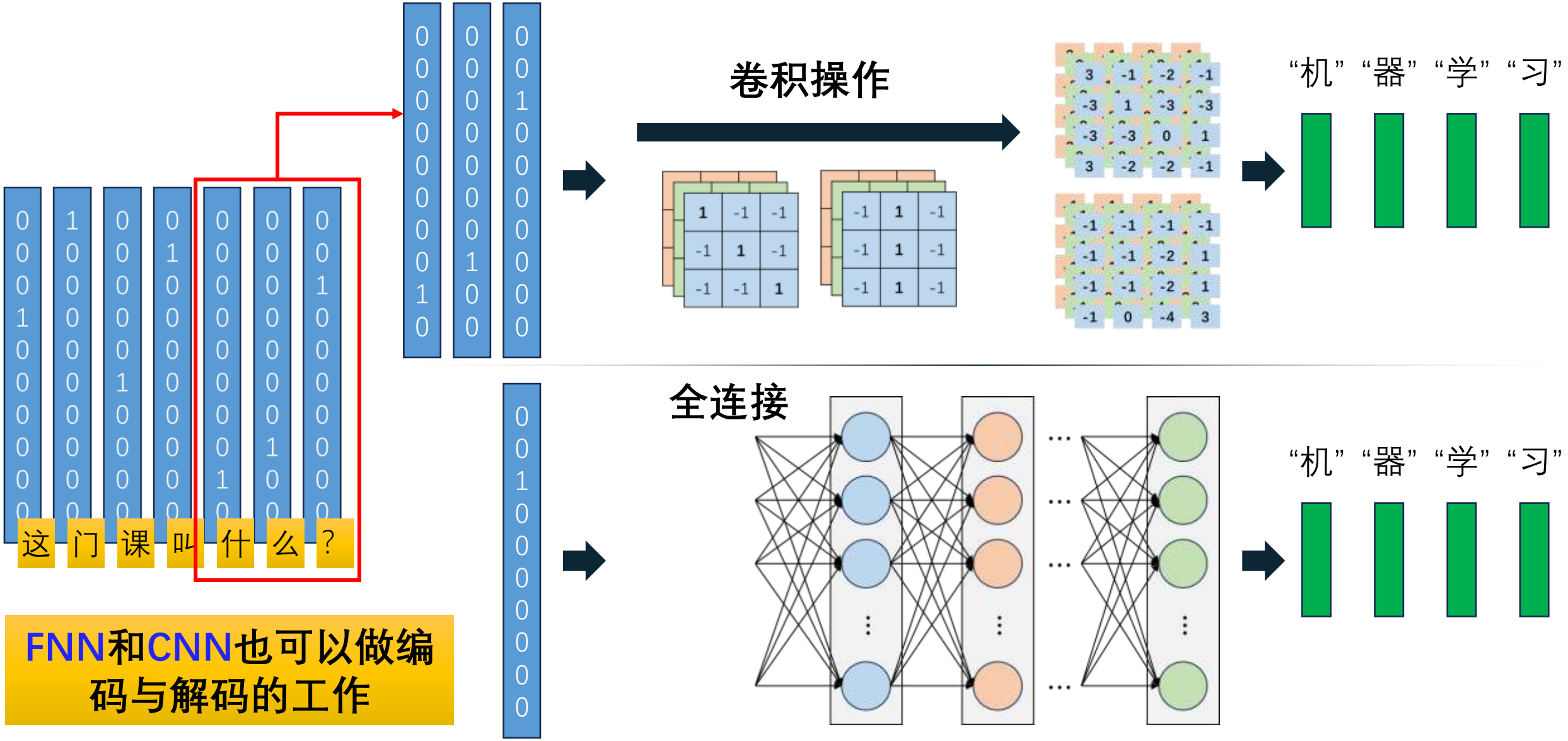
“问：这门课叫什么？机” “答：器”

“问：这门课叫什么？机器” “答：学”

“问：这门课叫什么？机器学” “答：习”

几乎所有涉及解码任务都可以转化为生成问题

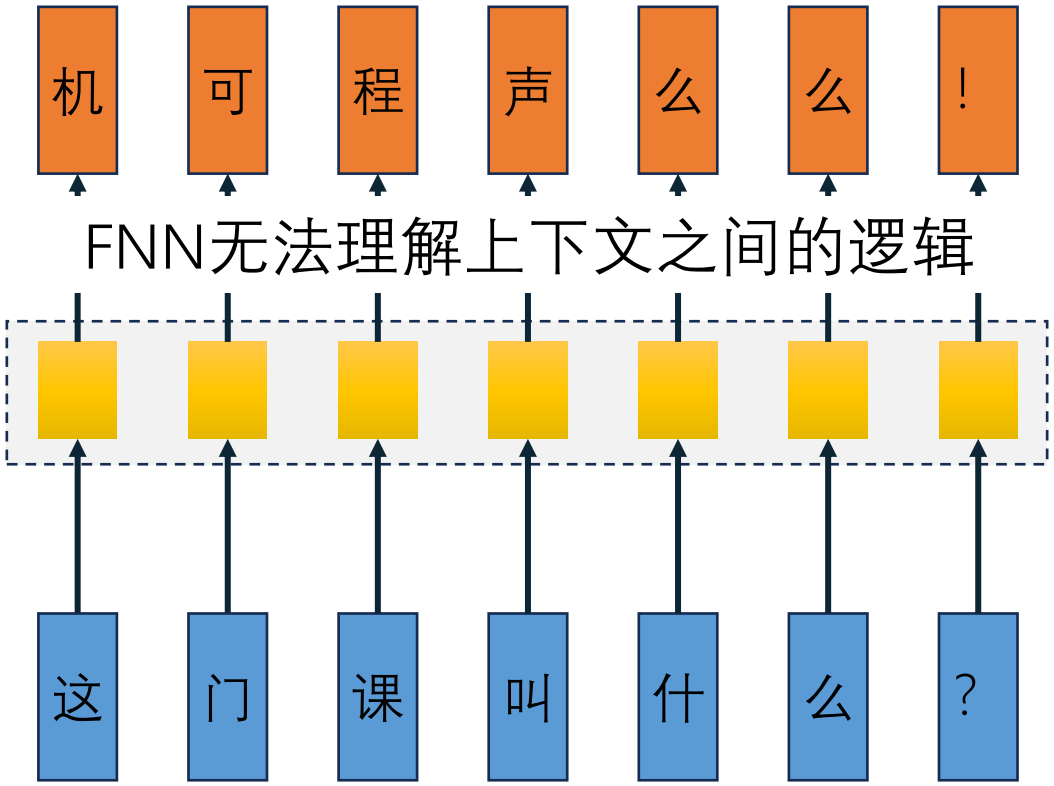
具体一点：可以用FNN和CNN进行解码吗？

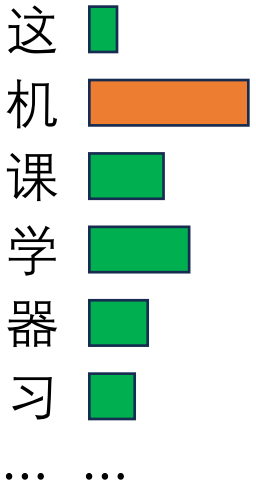


FNN和CNN也可以做编码与解码的工作

基于FNN的方法

FNN一次只能输入一个1-D向量





训练语料:

“这机器真稳定！”

“在这机器里面有个BUG！”

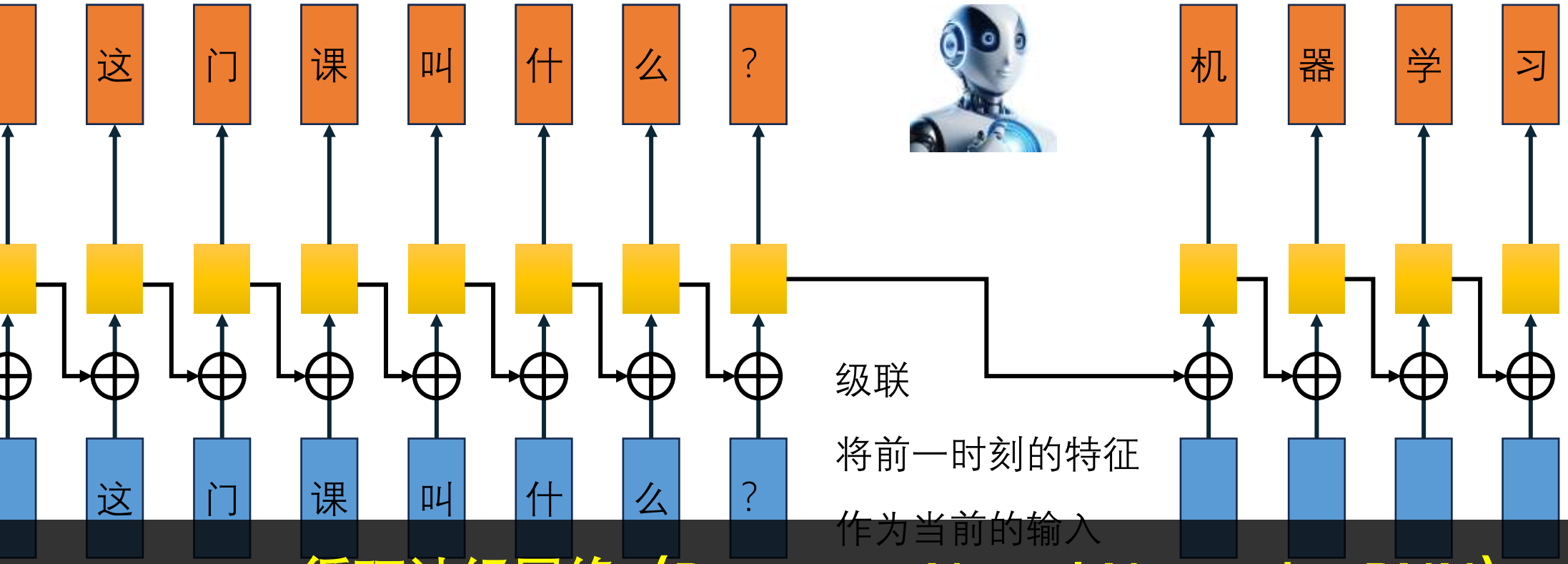
“这机会来之不易呀！”

“这门可真扎实！”

为了让输出的结果更稳定，
FNN应该学习向量之间在时间上的联系，
即上下文关系

由FNN到RNN

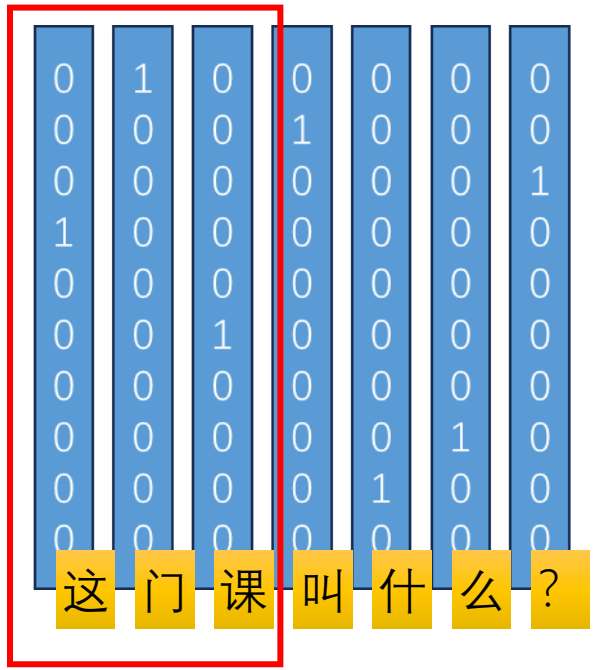
FNN一次只能输入一个1-D向量

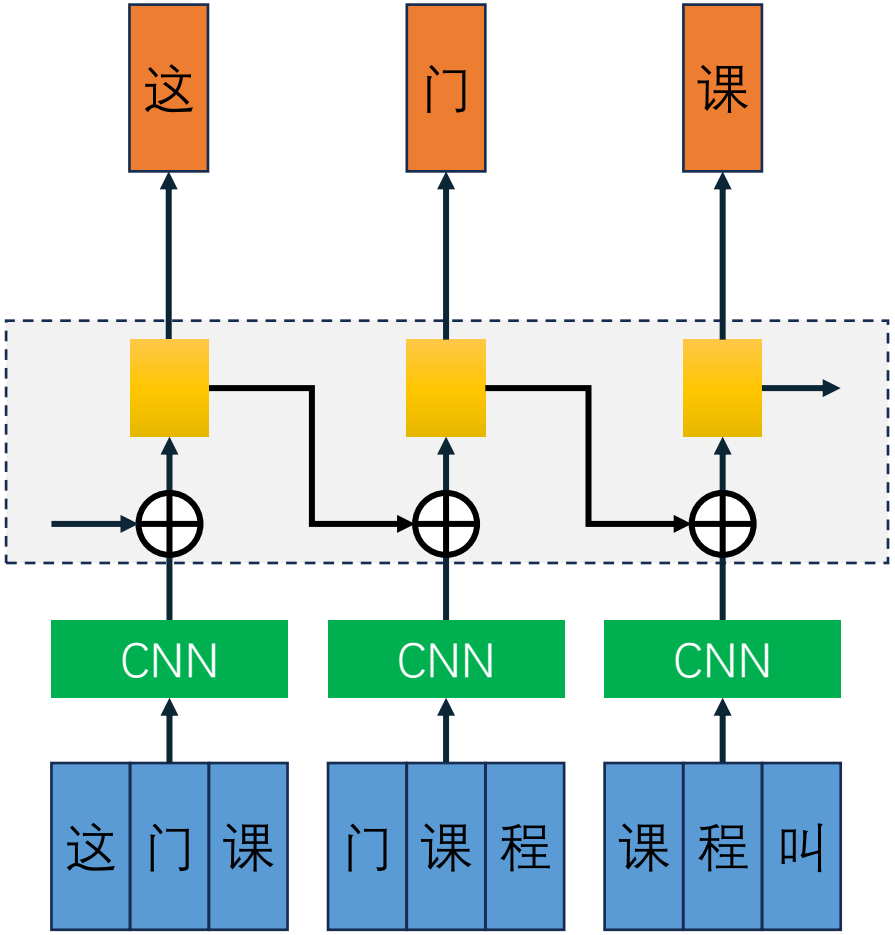


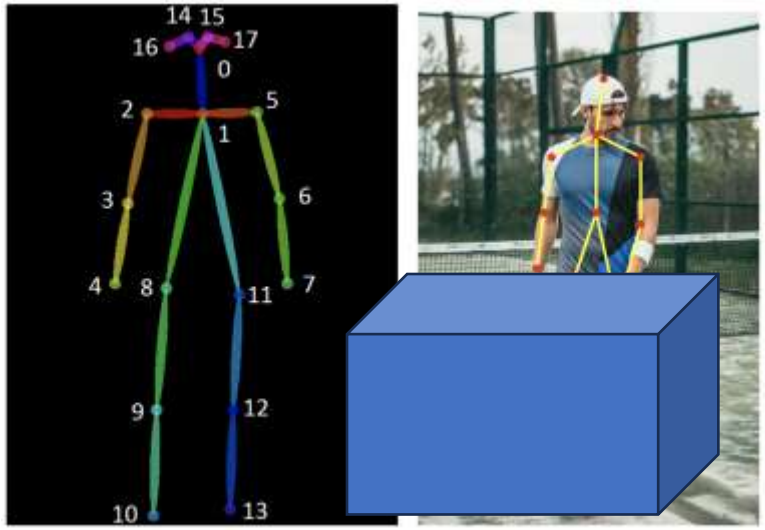
循环神经网络（Recurrent Neural Network，RNN）
具有记忆能力，能够学习时序数据的时间依赖关系

由CNN到CNN-RNN

CNN一次输入一个**矩阵**







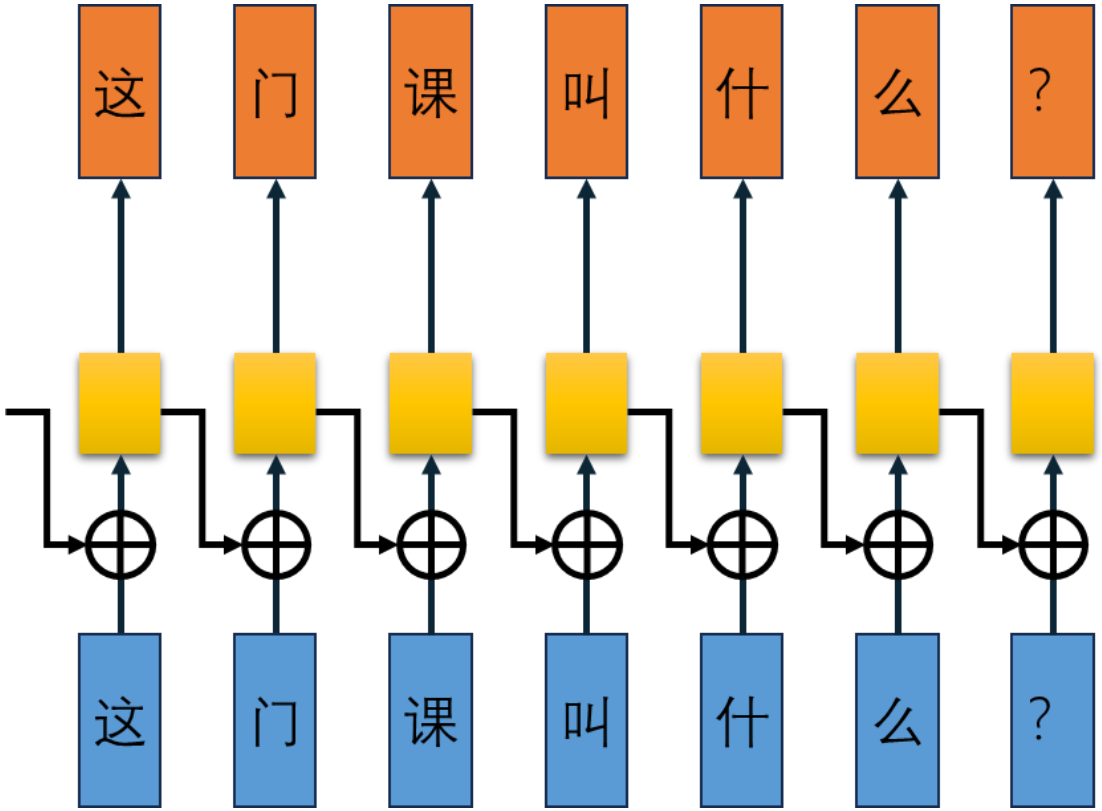
RNN

提取时间信息

合理推测/调整下一时刻的情况

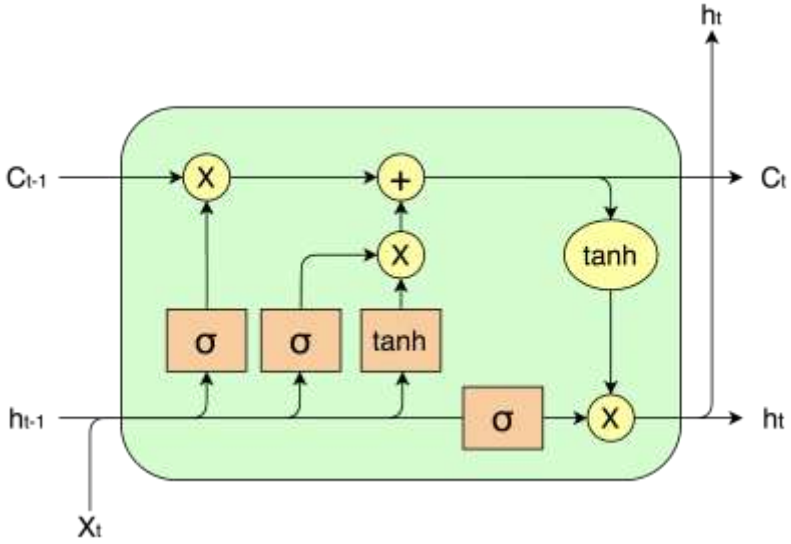
提取空间信息

RNN的不足以及LSTM



于尔根·施密德胡伯
Jürgen Schmidhuber

解决RNN没有“长期”记忆的能力



长短期记忆模型
Long Short-Term Memory,
LSTM

RNN仅具有短期记忆的能力，无法处理较长的时序

循环神经网络在机器翻译中的不足

English: *"The weather today is very nice."*

Target Chinese Translation: *"今天天气很好。"*

循环神经网络也许会翻译成……

"天气今天是非常好的"

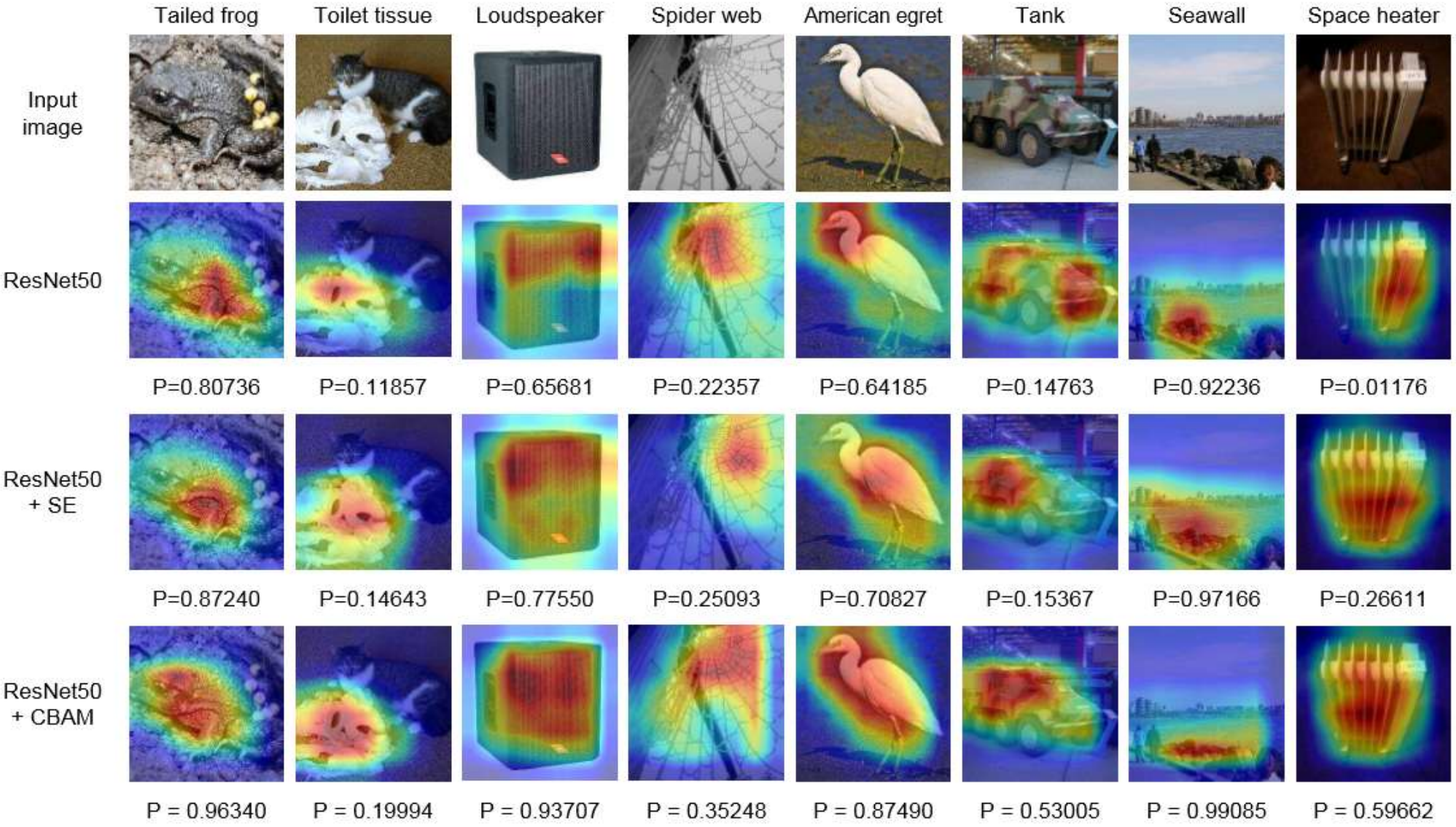
语序不当

不够地道

一眼机翻

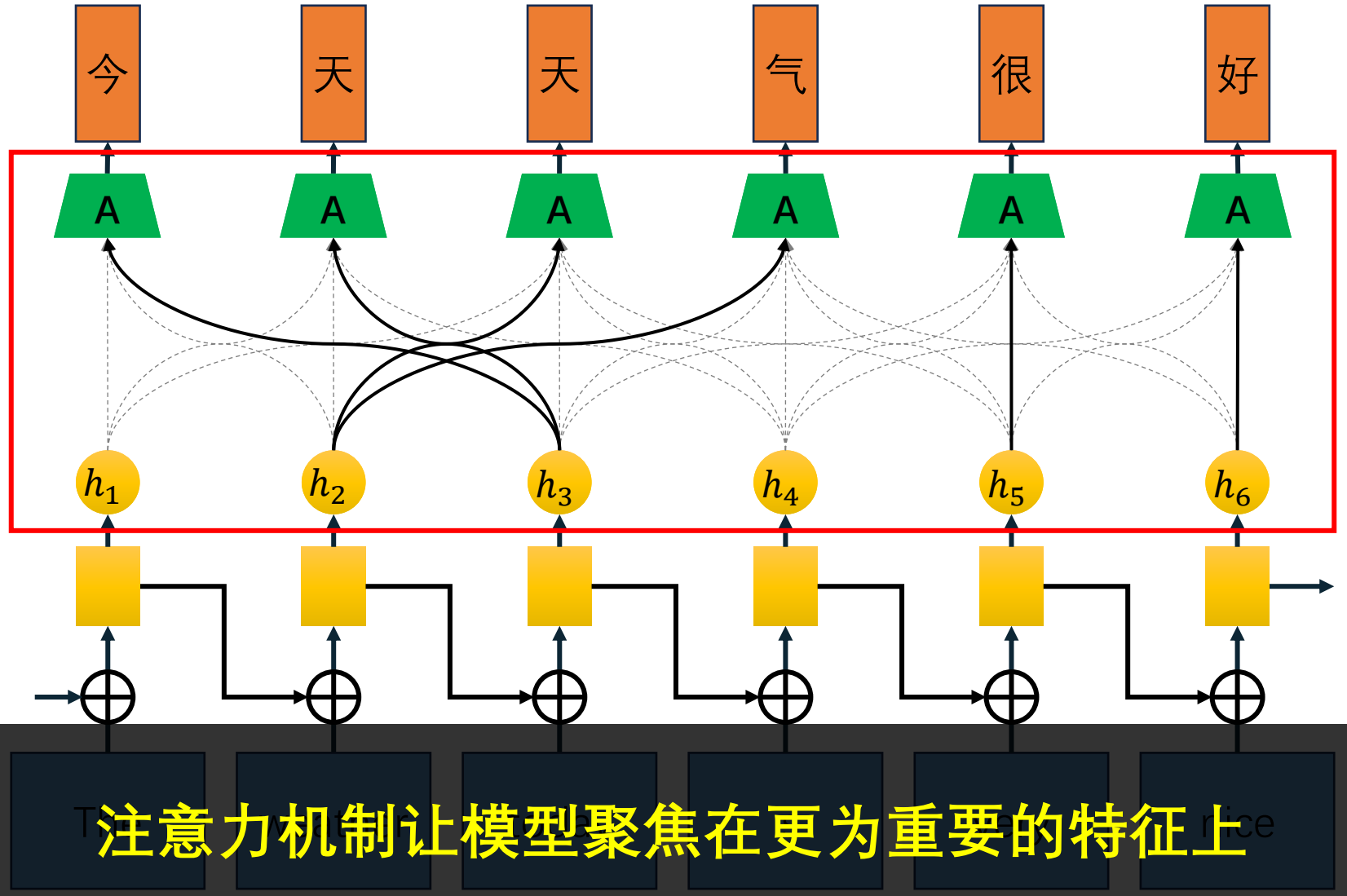
将**注意力机制**引入到NLP中
极大改善机器翻译的效果

注意力机制的直观理解


Tailed frog
Toilet tissue
Loudspeaker
Spider web
American egret
Tank
Seawall
Space heater
Input image
ResNet50
P=0.80736
P=0.11857
P=0.65681
P=0.22357
P=0.64185
P=0.14763
P=0.92236
P=0.01176
ResNet50 + SE
P=0.87240
P=0.14643
P=0.77550
P=0.25093
P=0.70827
P=0.15367
P=0.97166
P=0.26611
ResNet50 + CBAM
P = 0.96340
P = 0.19994
P = 0.93707
P = 0.35248
P = 0.87490
P = 0.53005
P = 0.99085
P = 0.59662

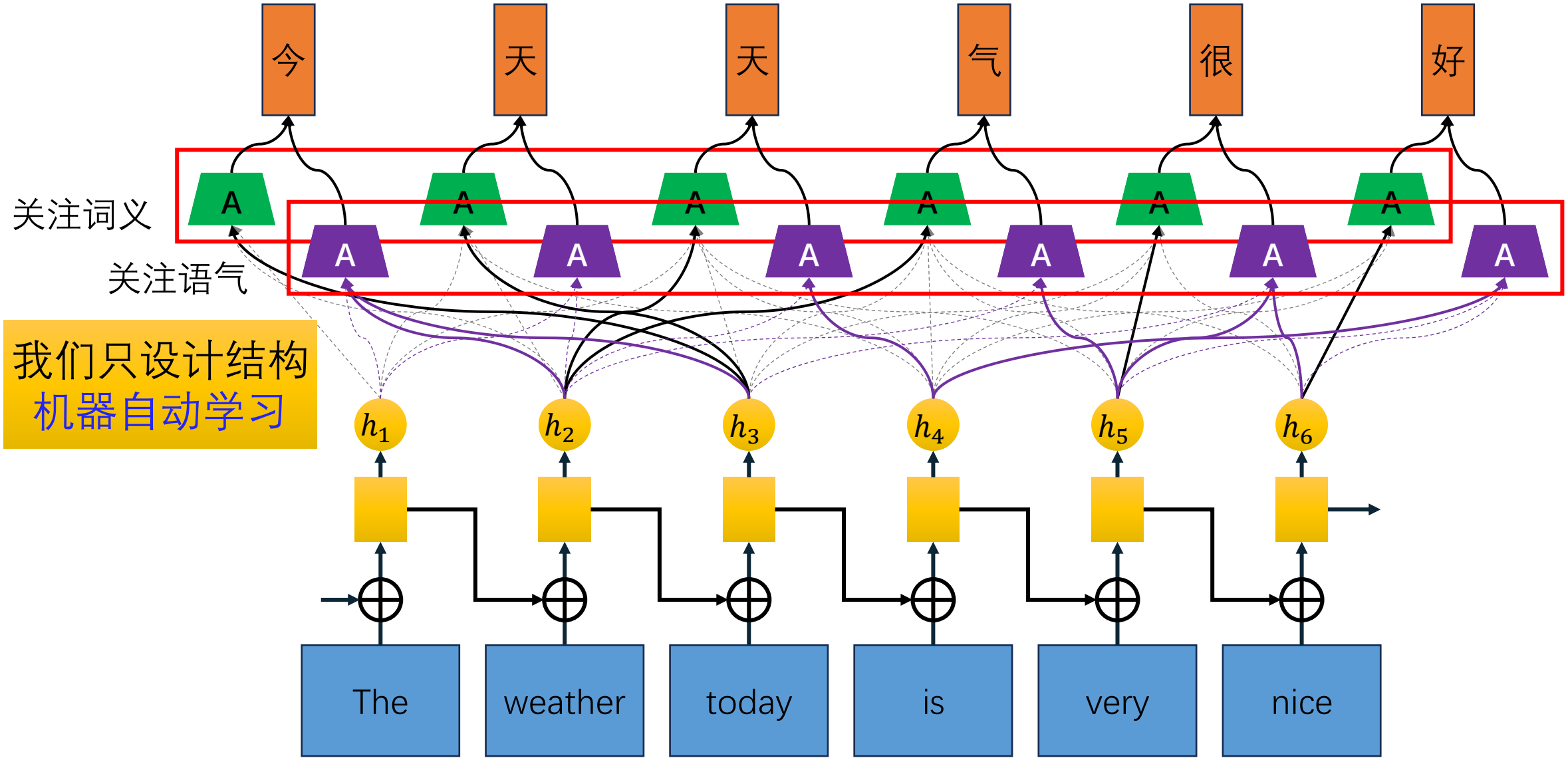
注意力机制

此部分较为复杂难以理解，所以记住前两页的图即可



注意力机制让模型聚焦在更为重要的特征上

注意力机制



我们只设计结构
机器自动学习

现在更受欢迎的是……

自注意力机制

Self-attention

Attention Is All You Need

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https://arxiv.org/abs/1706.03762